

*This exam has 2 pages and 8 exercises.*

*The result will be computed as (the total number of points plus 10) divided by 10.*

***Please motivate all answers!***

By participating in this exam, I declare to understand that taking an online exam during this corona crisis is an emergency measure to prevent study delays as much as possible. I know that fraud control will be tightened and realize that a special appeal is being made to trust my integrity. With this statement, I promise to:

- make this exam completely on my own, and not share my solutions with other students,
- not consult sources other than explicitly mentioned in the examination, and
- make myself available for any oral explanation of my answers.

**1. Distributivity laws** (6 + 6 points)

For each of the two distributivity laws below, regarding conjunction and exclusive or, either argue that it holds or give a counterexample to show that it does not hold.

(a)  $(x \wedge y) \oplus z \equiv (x \oplus z) \wedge (y \oplus z)$

(b)  $(x \oplus y) \wedge z \equiv (x \wedge z) \oplus (y \wedge z)$

**2. Logic circuit** (10 points)

Give a logic circuit, containing only AND, OR and NOT gates, that captures the formula  $(y \rightarrow x) \rightarrow \neg(x \rightarrow z)$ .

**3. DNF + axioms for  $\equiv$**  (4 + 7 points)

- (a) Provide the truth table for the formula  $p \vee q$ , and extract a DNF from this truth table that is semantically equivalent to  $p \vee q$ .
- (b) Derive with the axioms for semantic equivalence that  $p \vee q$  and the DNF you extracted in (a) are semantically equivalent.

**4. Predicate logic** (12 points)

Give two models to show that the following two formulas do not semantically entail each other.

- $\exists x C(x)$
- $\forall x C(x)$

**5. Relations** (6 + 4 points)

- (a) Is the empty binary relation  $R_1$  in  $\mathbb{N}$ , which relates no pairs of elements, reflexive? Is it transitive? Is it symmetric? Is it anti-symmetric?
- Is the identity binary relation  $R_2$  in  $\mathbb{N}$ , which relates each element only to itself, reflexive? Is it transitive? Is it symmetric? Is it anti-symmetric?
- (b) Describe the binary relations  $R_1 \circ S$  and  $R_2 \circ S$ , for any binary relation  $S$  in  $\mathbb{N}$ .

**6. Sets** (5 + 7.5 points)

Consider the following set-theoretic equality:

$$(A \setminus B)' \cap (A \setminus C)' = A' \cup (B \cap C)$$

- (a) Draw Venn diagrams for the two sets  $(A \setminus B)' \cap (A \setminus C)'$  and  $A' \cup (B \cap C)$ , depicting clearly which area corresponds to the set, and how the set is constructed from  $A$ ,  $B$  and  $C$  using the fundamental set operations. Use your Venn diagrams to conclude that the set-theoretic equality displayed above holds.
- (b) Prove the set-theoretic equality displayed above (for all sets  $A$ ,  $B$  and  $C$ ) using the rules of the algebra of sets.

**7. Functions** (6 + 4 points)

We are given the functions  $f: \mathbb{R} \rightarrow \mathbb{R}$ ,  $mul_3: \mathbb{R} \rightarrow \mathbb{R}$ , and  $g: \mathbb{R} \rightarrow \mathbb{R}$  defined by

$$f(x) := \frac{1}{x^2 - 1} \quad mul_3(x) := 3x \quad g(x) := \sqrt{x - 1}$$

- (a) What is the domain of definition of  $f$ , and what is its range?
- What is the domain of definition of  $g$ , and what is its range?
- (b) We define  $h = f \circ mul_3^{-1} \circ g$ . Give an explicit formula for  $h(x)$ .

**8. Induction** (12.5 points)

Prove using mathematical induction that the following statement is true for all integer numbers  $n \geq 1$ :

$$\sum_{k=1}^n (k-1)^3 = \frac{((n-1)n)^2}{4}$$